Ring Modulator Layout Program 2

By Dr.-Ing. Marc Schneider from the Institute for Data Processing and Electronics (IPE) of the Karlsruhe Institute of Technology (KIT), Germany.

Program version: 0.2

Documentation: 2026-08-17

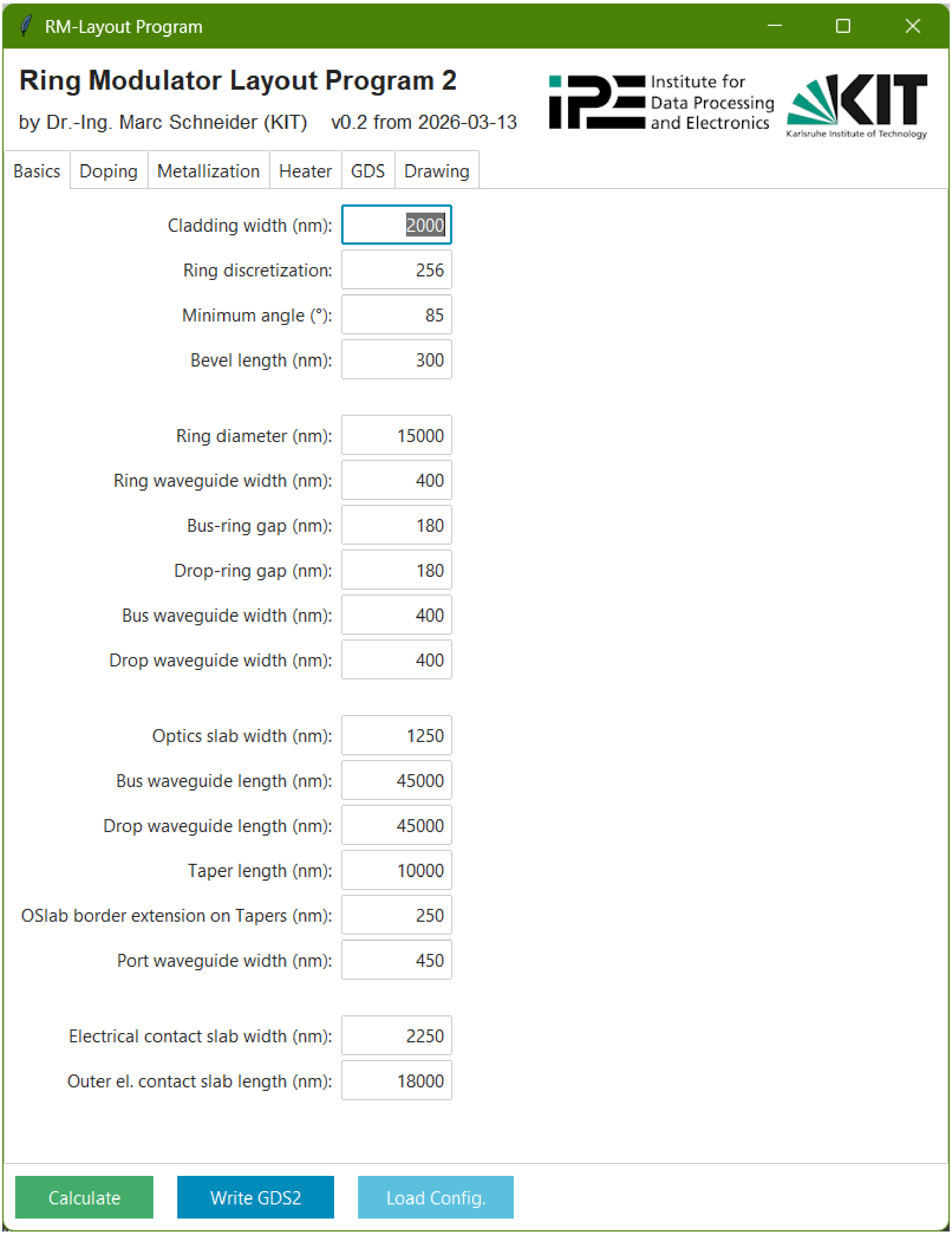
This program is for designing silicon photonic ring modulators (RM) with a pn-junction along the circular ring to modulate light traversing through a loosely coupled, straight bus waveguide. The result can then be saved as GDSII file for use in your photonic chip design.

In resonance of the ring, light can be coupled to a straight drop waveguide, whose distance to the ring and therefore coupling parameter can be adjusted independently from the distance of the bus waveguide to the ring. The outer waveguide connections are designed as strip waveguides, which taper to rib waveguides with slabs in the whole RM region. The slab thickness of the bus and drop waveguides, as well as the outer ring slab is defined by the GDS layer for the dedicated etch depth. The slab thickness of the inner ring slab can be chosen as the same or a different GDS layer with a different etch depth. To change the resonance of the ring through the thermo-optic effect, a heater on top of the ring is implemented.

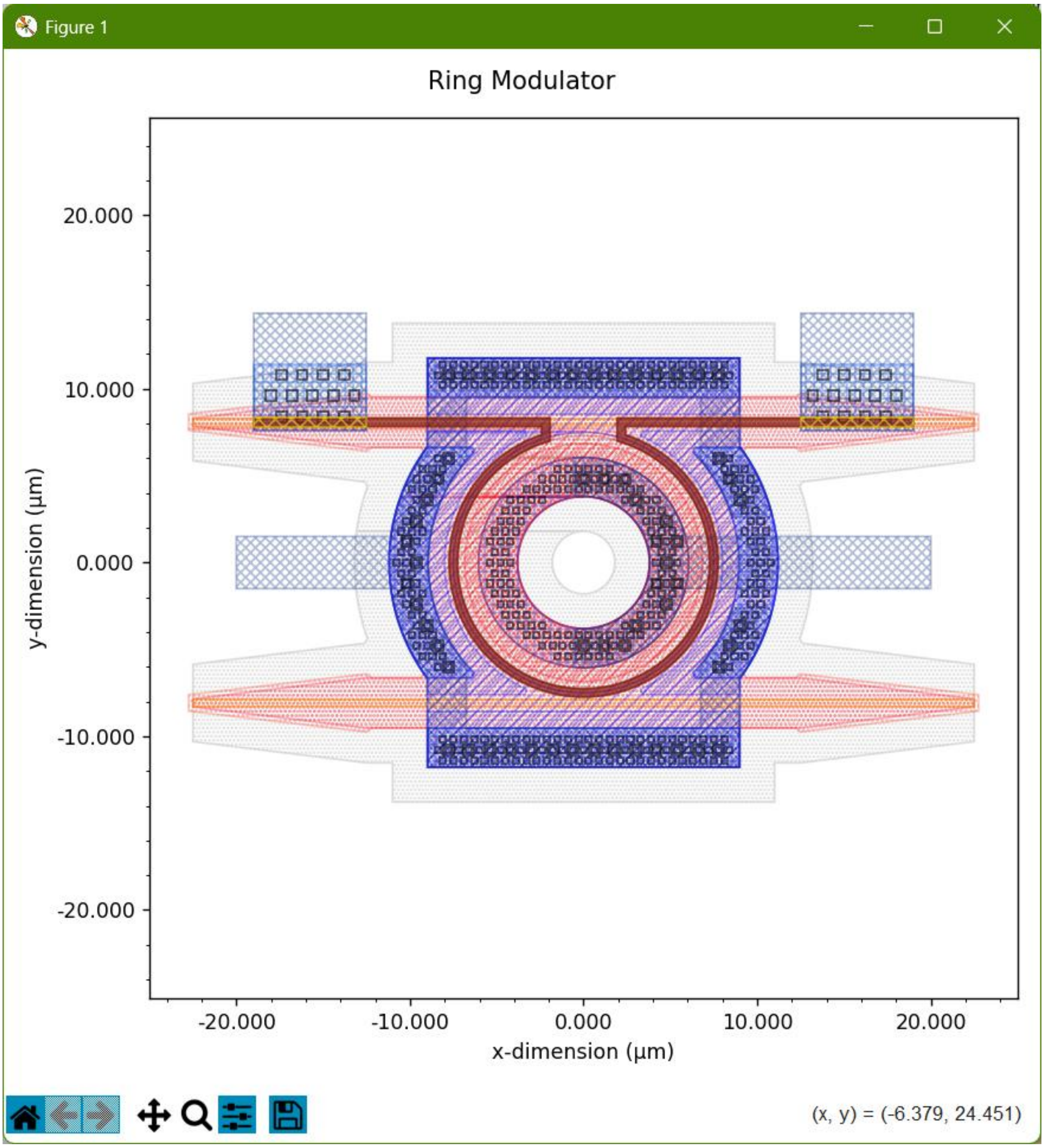
Some geometry aspects are influenced by the design rules of a certain European fab, but as the design is widely adjustable, it should be usable for many different fabs.

The program is written in Python (my first Python program, so you might notice some awkward constructions) and depends heavily on the gdsfactory library (https://github.com/gdsfactory/gdsfactory).

After program start, a single window appears with some program information, six tabs for parameters and three buttons at the bottom. The ‘Basics’ tab is selected.



Here, basic parameters can be entered. Pressing the "Calculate" button lets the program calculate the full geometry of a ring modulator with bus and drop waveguides and shows a new window with the result:



You can pan and zoom into the geometry and also save a picture of the figure.

If you are satisfied, you can save a GDSII-file with the layout by pressing the “Write GDS2” button in the main window. You’ll have to enter a valid file name in the file dialog popping up.

Besides the GDSII file, two additional files will be generated. One is a small Python function which loads the GDSII file and adds port definitions for use in a chip layout made with Python and gdsfactory. The other file contains the configuration of the saved layout with all parameters. Such a file can be reused by loading it with the button “Load Config.”. It’s a plain text file and can also be edited by a standard text editor. If you edit the file and delete parameters, those parameters are not changed from those defined in the program GUI. This is especially helpful, if you have a certain GDS layer configuration and don’t want to change the Python program. Adjust the layers in the GDS tab, write a GDS file, delete all entries in the [Variables] section except the GDS_DBU and GDS_UU parameters, and save the file to a new name. If you load this file into the RM Layout Program, only the parameters of the GDS tab will be adjusted.

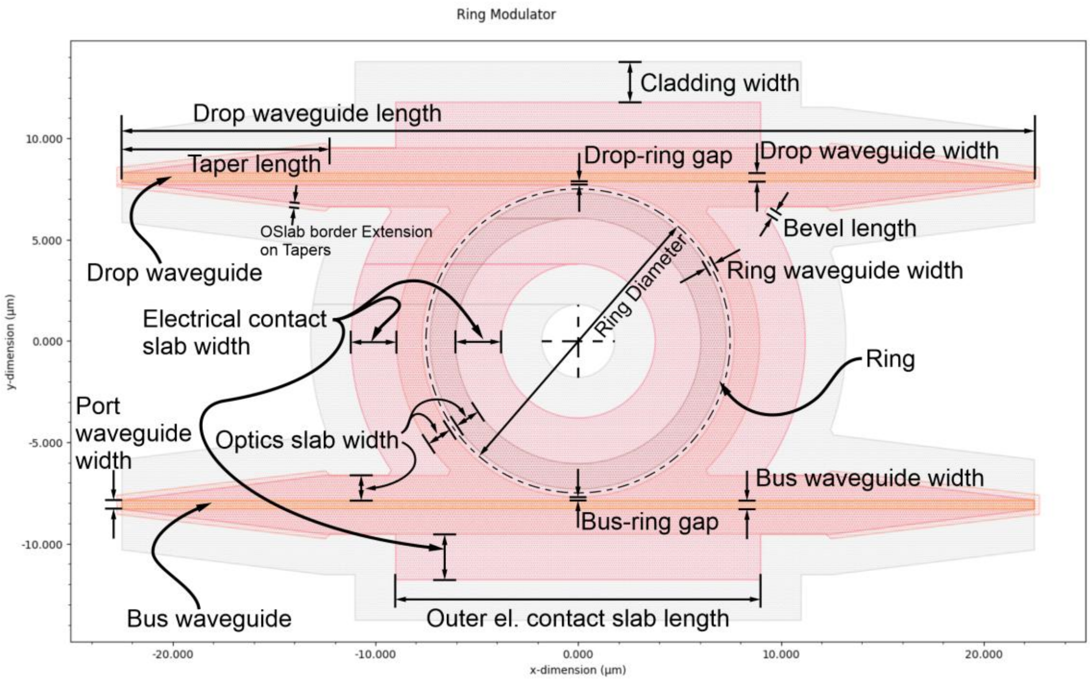
Now let's discuss all parameters in the different tabs.

Basics tab:

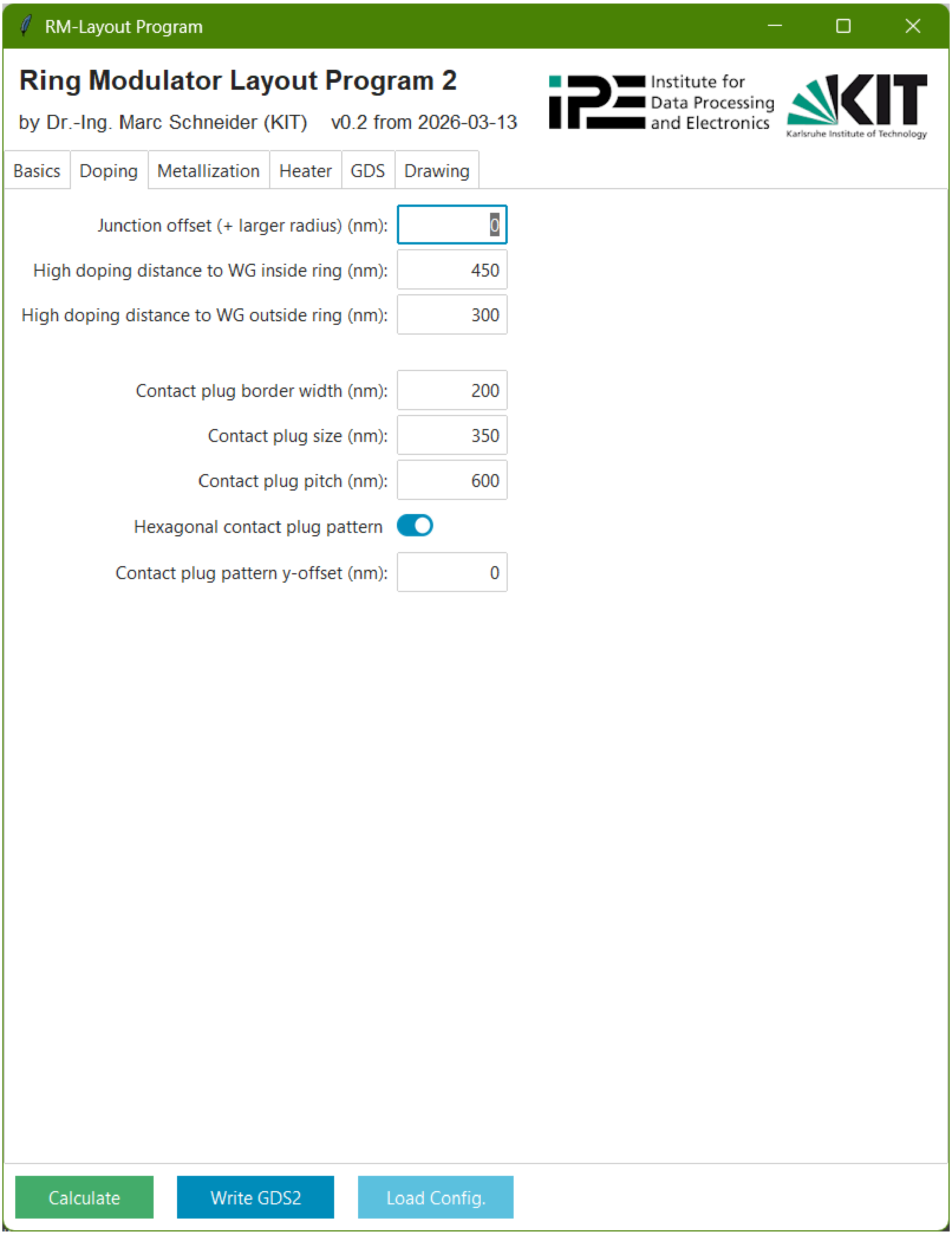
Cladding width	Cladding width around structures in nm. In general, silicon photonic components are etched out of SOI wafers. To separate the RM from the rest of the wafer, the optical structures are surrounded by cladding of the defined width.
Ring discretization	Number of vertices in full circle of ring. GDSII only knows straight lines so that curves have to be split into straight line segments. This parameter defines, how many vertices are used for the discretization of a full circle. The standard value of 256 divides the 360° of a circle into 256 slices, so vertices are placed each 360°/256 ≈ 1.36°.
Minimum angle	Minimum 'steep' angle for structure in degrees. At some places, sharp, pointy angles occur in the design, but some design rules forbid those angles. Therefor corners with an angle smaller than or equal the given one are beveled.
Bevel length	Length of bevels to mitigate sharp angels (nm). If a sharp angle gets a bevel, the bevel has to be of a certain length to prevent minimum feature size design rules from hitting. The given value defines the minimum length of such a bevel, in some cases it might become slightly longer.
Ring diameter	Diameter of the center line of the ring waveguide in nm.
Ring waveguide width	Width of the rib of the ring waveguide in nm.
Bus gap	Gap between ring and bus waveguide in nm. The gap between the rib of the bus waveguide and the rib of the ring waveguide. How deep the gap is, is determined by the appropriate GDS layer for the outer optical slab cladding.
Drop gap	Gap between ring and drop waveguide in nm. Same as for the bus gap.
Bus waveguide width	Width of bus waveguide in nm, relevant for the coupling section to the ring.
Drop waveguide width	Width of drop waveguide in nm, relevant for the coupling section to the ring.
Optics slab width	Width of optical slabs (thin etched regions) around rib waveguides in nm. Thinner slabs adjacent to ribs, important for light guidance and should be wide enough that no light leaks into the electrical contact slab.

Bus waveguide length	Length of the whole bus waveguide from port to port.
Drop waveguide length	Length of the whole drop waveguide from port to port.
Taper length	Length of tapers. Within this length the bus and drop waveguides transform from a strip waveguide at the ports to the rib waveguides used in the component. At the same time, the rib width transforms from the port waveguide width to the bus/drop waveguide width.
OSlab border extension on Tapers	Tapers of optical slabs get a border of this size to avoid sharp angles at the ports for the slab definition (required by some design rules).
Port waveguide width	Width of the waveguide strip at the ports in nm.
Electrical contact slab width	Width of electrical contact slabs (non-etched regions) around optical slabs in nm. In the ring region electrical contacts for the pn junction are needed and this parameter defines, how wide the contacts are.
Outer el. Contact slab length	Length of outer electrical slabs. The outer electrical contact is divided into four sections, two semi-circular to the left and to the right of the ring and two at the top and at the bottom, beyond the bus and the drop waveguides. This parameter defines the length of the latter sections and have to be adjusted to get nice contacts.

This picture shows all these parameters in the layout:

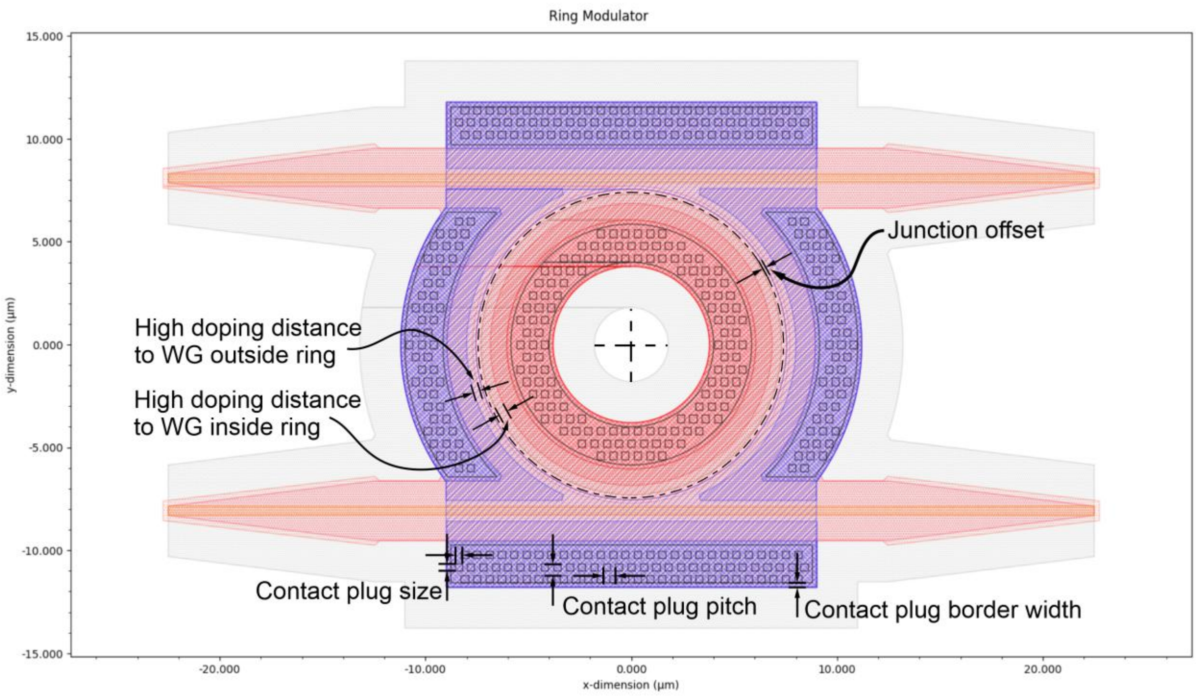


Doping tab:



Junction offset	Offset position of circular pn-junction from center line of ring rib. Positive values shift the junction line to the outside, negative to the inside.

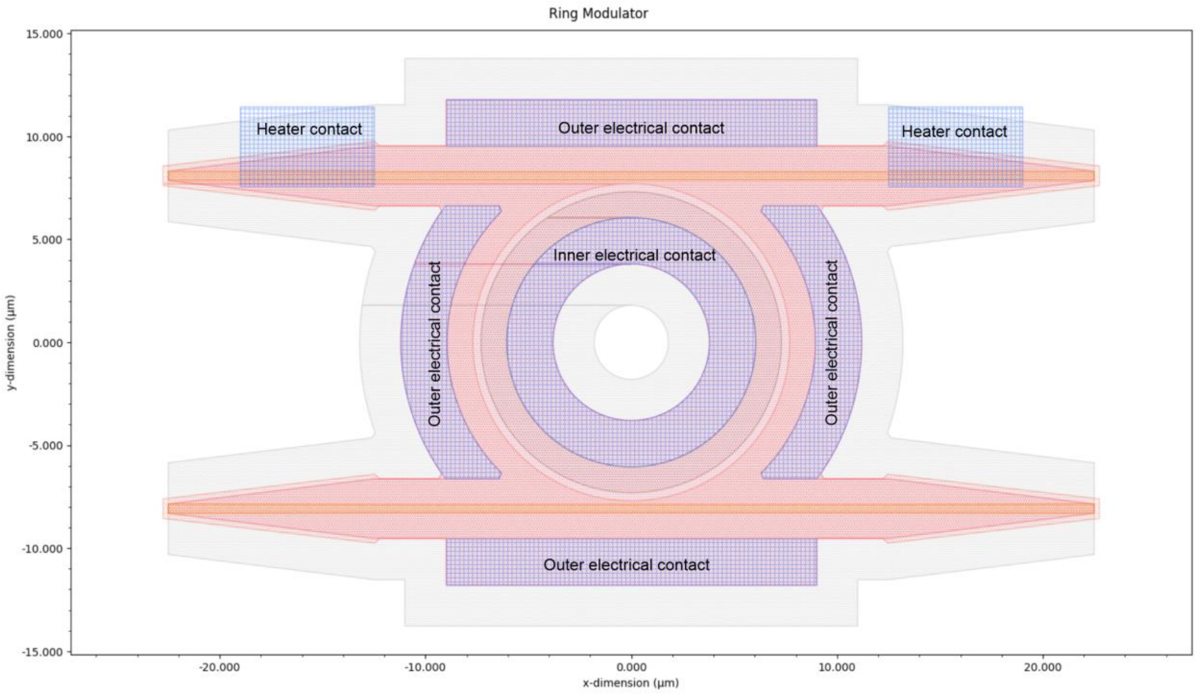
High doping distance to WG inside ring	Distance between ring waveguide rib border and start of high doping region inside the ring. Distance should be large enough that no relevant parts of the optical mode gets into the high doping region.
High doping distance to WG outside ring	Distance between ring waveguide rib border and start of high doping region outside the ring. Distance should be large enough that no relevant parts of the optical mode gets into the high doping region.
Contact plug border width	The whole electrical contact slab area will be doped with contact doping. To connect this area to the metal 1 layer, square contact plugs are used. Maybe these contact plugs are only allowed inside a certain distance to the border of the doped area and this parameter defines the distance of the plugs to the contact doping border.
Contact plug size	Width and height of silicon contact plug square in nm.
Contact plug pitch	Horizontal and vertical pitch (center to center distance) between silicon contact plugs in nm.
Hexagonal contact plug pattern	Switch to choose between a square or a hexagonal contact plug pattern.
Contact plug pattern y-offset	The contact plug pattern can be shifted vertically to sometimes achieve a better covering of contact plugs inside the allowed regions. Positive values shift the pattern upwards, negative values downwards.



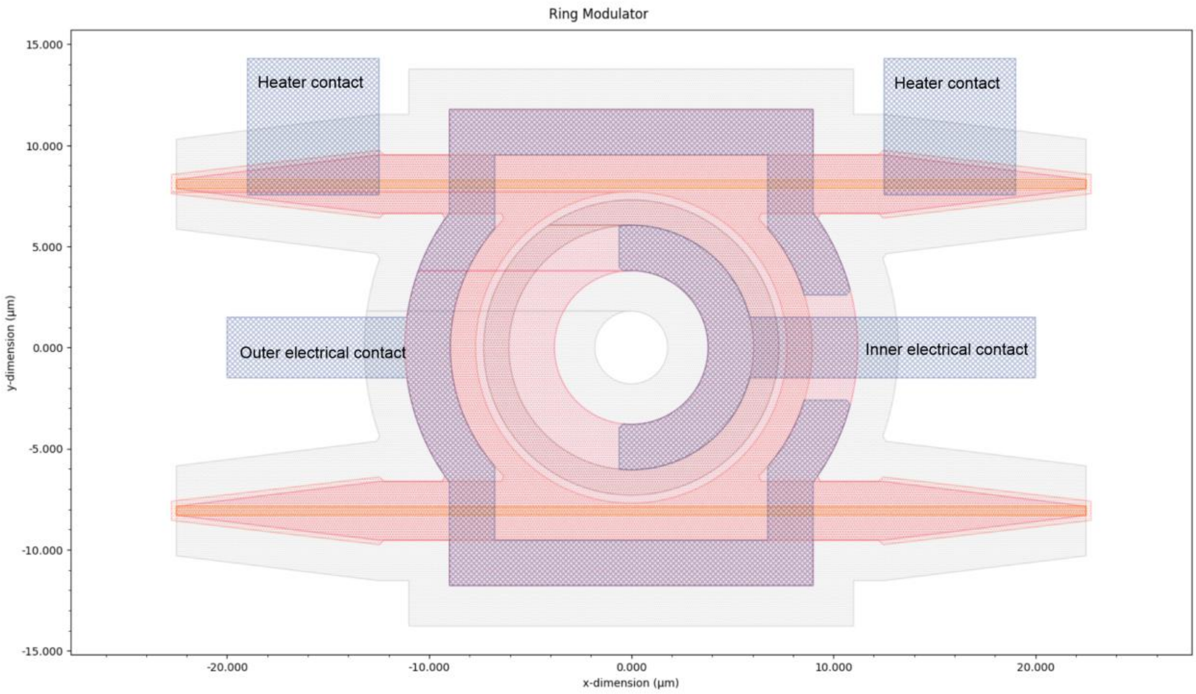
Metallization tab:

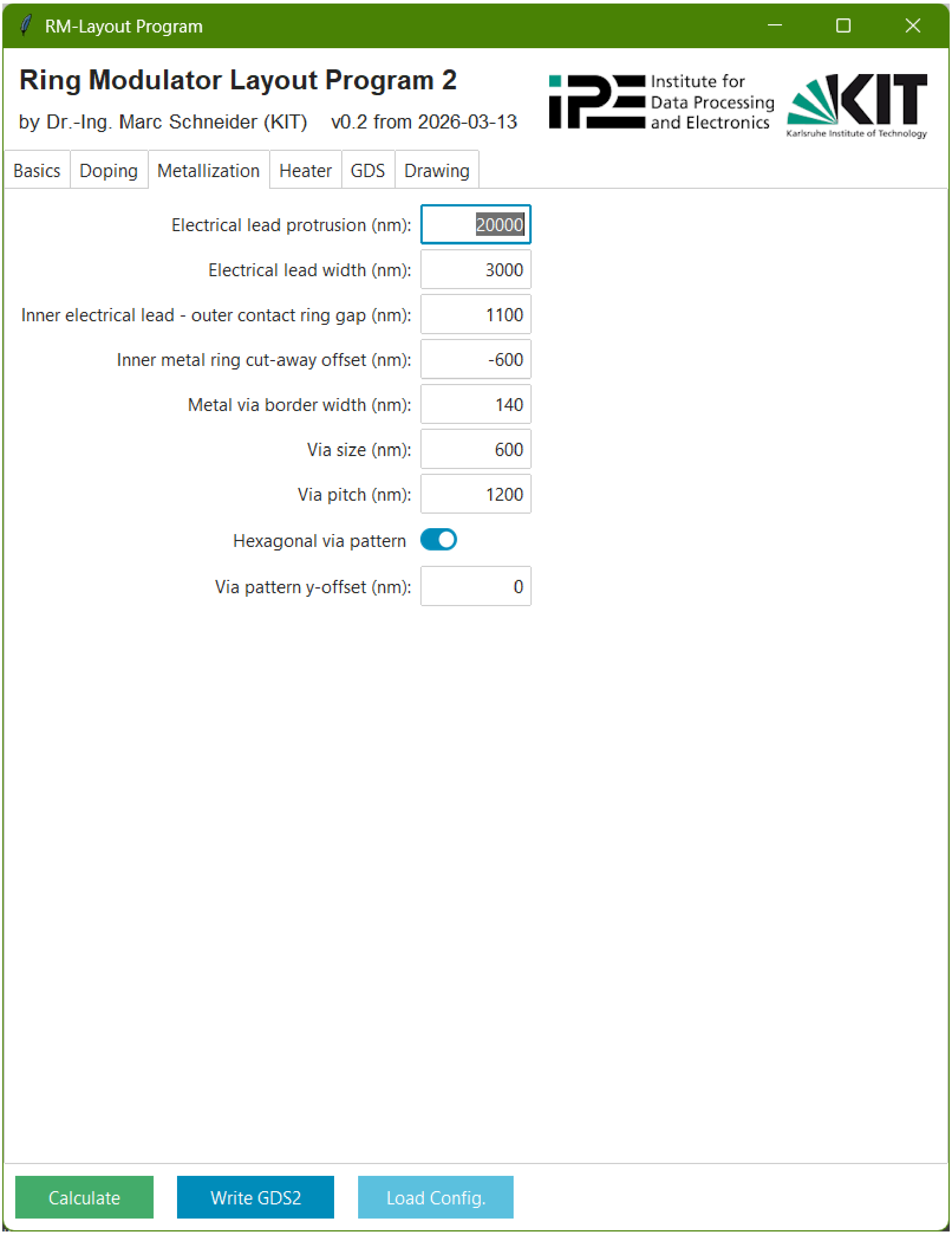
Metallization is done with two layers, Metal 1 and Metal 2 with vias connecting both.

Metal 1 covers the areas of the outer and inner electrical contact slabs, as well as two rectangular areas at the top left and right for the heater:



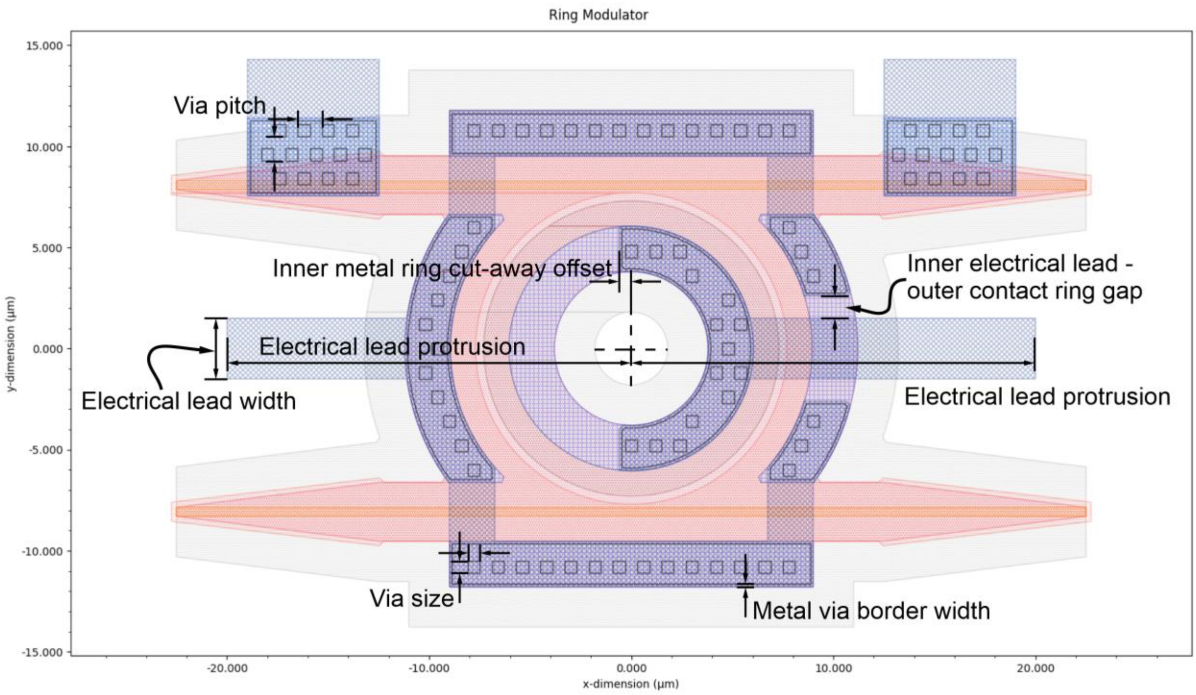
Metal 2 connects the inner Metal 1 ring and routs it to the right to an electrical port, as well as connects the four outer Metal 1 areas and adds an electrical port on the left side. Additionally the heater contacts are routed to the same metal 2 layer and extended:





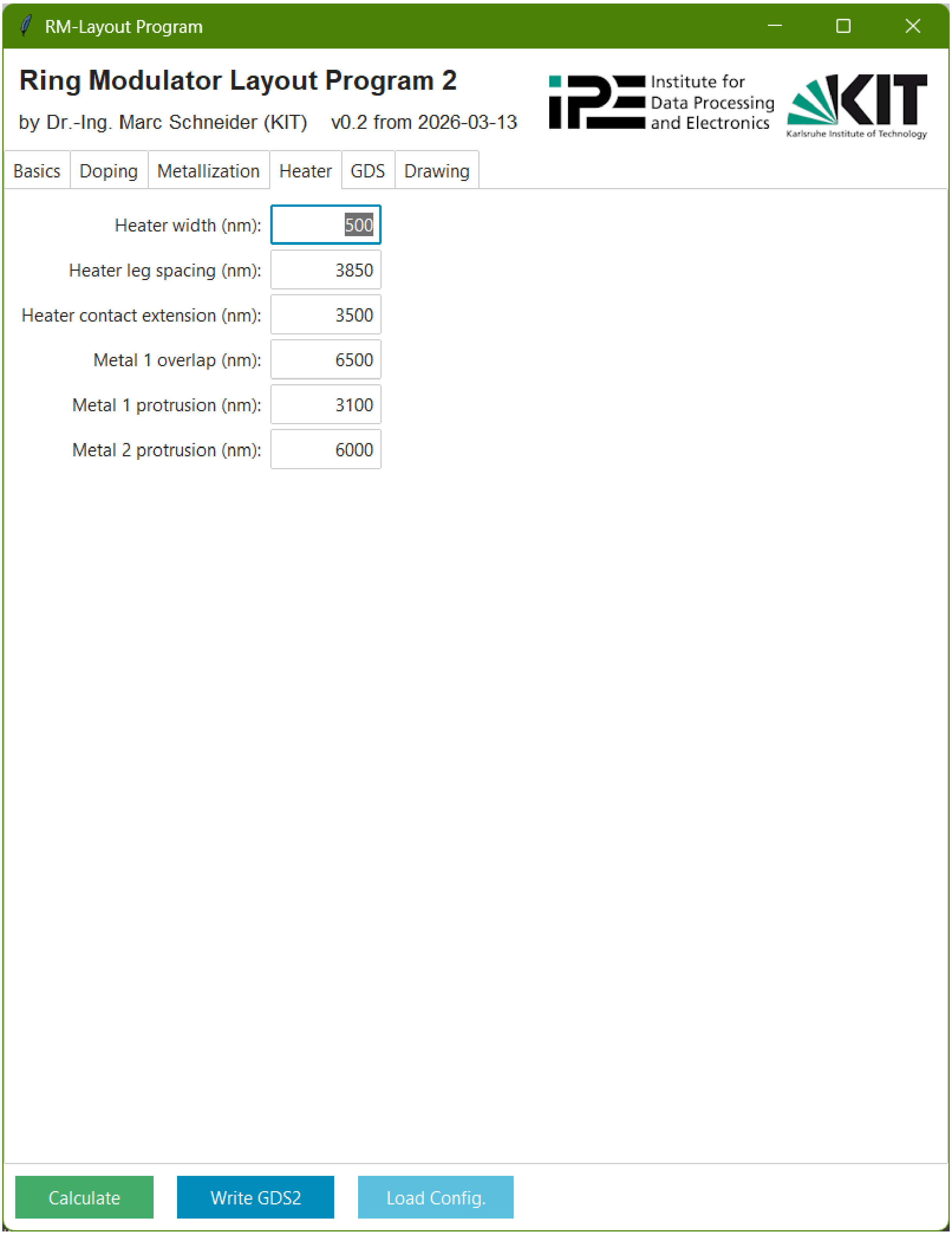
Electrical lead protrusion	Horizontal distance between center (0|0) and outer edge of metal leads of RF contacts. With the given value of 20000 nm the left electrical port for the outer contact is at (-20000|0) and the right electrical port for the inner contact is at (+20000|0).

Electrical lead width	Width of electrical RF port metal leads.
Inner electrical lead – outer contact ring gap	Gap between inner electrical RF port lead on the right and metal 2 connector of outer electrical contact.
Inner metal ring cut-away offset	Metal 2 for the inner contact doesn't form a complete, but only a half ring. The cut plane orients at y=0 nm, but can be offset horizontally to make room to add some more vias.
Metal via border width	The whole electrical contact slab area will be doped with contact doping and the metal 1 regions have the same shape and dimensions as these areas. Metal 2 has different shapes and the overlaps should be connected by square vias. Maybe these vias are only allowed inside a certain distance to the border of the metallizations and this parameter defines the distance of the vias to the metallization edges.
Via size	Width and height of via square.
Via pitch	Horizontal and vertical pitch (center to center distance) between vias.
Hexagonal via pattern	Switch to choose between a square or a hexagonal via pattern.
Via pattern y-offset	The via pattern can be shifted vertically to sometimes achieve a better covering of vias inside the allowed regions. Positive values shift the pattern upwards, negative values downwards.

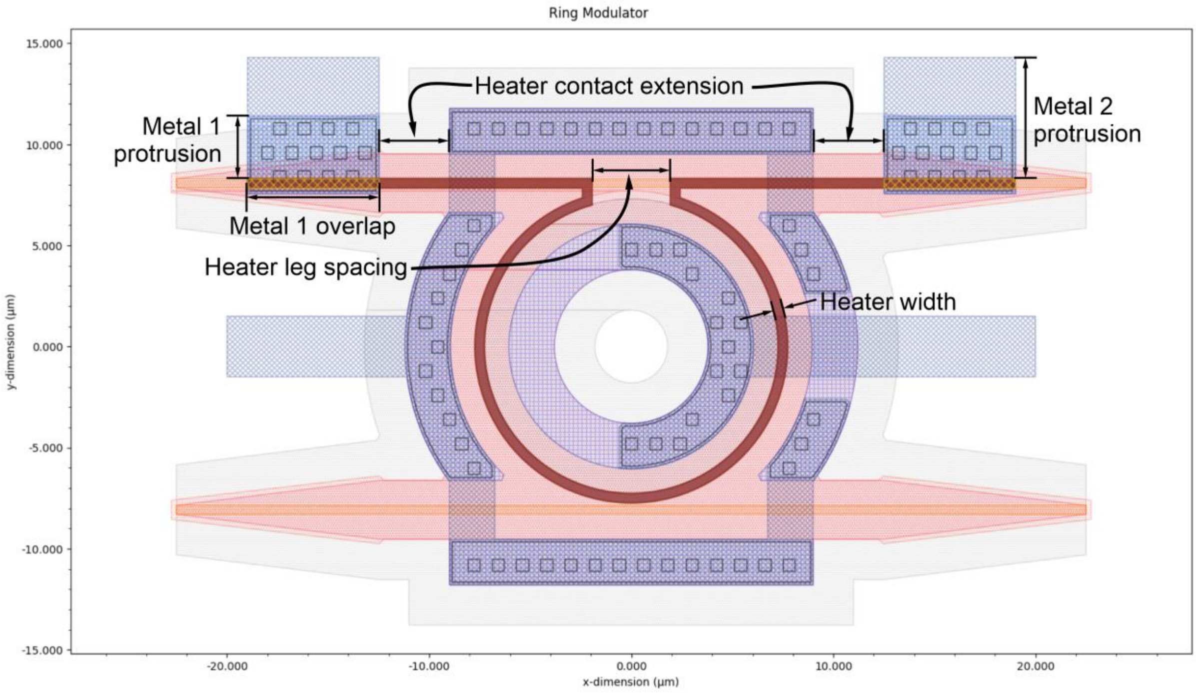


Heater tab:

The heater is an omega shaped structure on top of the ring waveguide with connections to the outside on top of the drop waveguide. The connection between the heater structure and metal 1 is marked in the GDS layer “Heater Contact”.

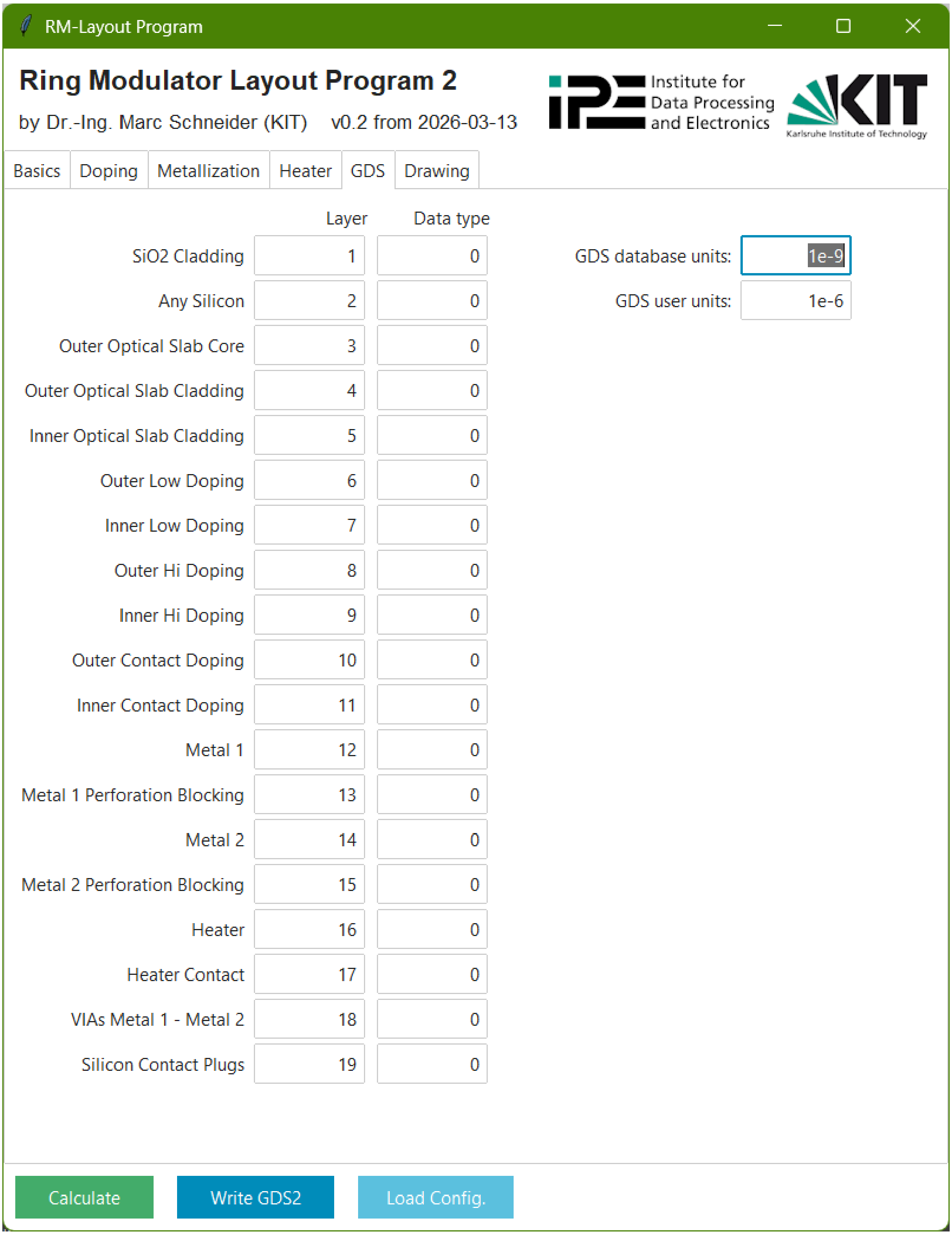


Heater width	Width of heater structure.
Heater leg spacing	Spacing between the heater lines at the narrow part of the omega shape.
Heater contact extension	Distance between contacting leads of heater and the metallization of the outer electrical slab on the top of the device. (I know that the name sucks…)
Metal 1 overlap	Overlap between heater structure and metal 1 for contacting. Defines the width of the electrical contacts at the heater contact ports.
Metal 1 protrusion	Protrusion of metal 1 over heater structure in +y direction. Just important to fit some vias into the area.
Metal 2 protrusion	Protrusion of metal 2 over heater structure in +y direction. This defines the y-position of the heater contact ports.



GDS tab:

In this tab the different GDS layers with their data types are defined as well as the GDS database and user units. All entries should be defined in the PDK of the intended process.



GDS database units	Units for GDS database. Most often 1 nm, which is 1e-9.
GDS user units	Units for GDS geometry. Most often 1 µm, which is 1e-6.
SiO2 Cladding	Silicon dioxide cladding surrounding the device
Any Silicon	If any silicon of the SOI waver should remain it is marked in this layer. Could be full height or etched to any depth, but never down to the SiO2 layer. If nothing else is defined, it should be full height, as this is how the rib of the ring waveguide is defined.
Outer Optical Slab Core	This is the slab region at the outside of the ring. To define bus and drop waveguide with full height, they are defined in this layer. In process logic it means that the bus drop waveguide is etched any silicon, nominally etched down to a certain level, but not, so that the full height remains. The design rules of a certain fab demand this procedure…
Outer Optical Slab Cladding	This is the slab region at the outside of the ring and besides bus and drop waveguides. The PDK defines, how deep these regions are etched.
Inner Optical Slab Cladding	This is the slab region at the inside of the ring. The PDK defines, how deep these regions are etched and as this can be different from the outside, you can build a ring waveguide cross section with a thin slab at the outside for good light confinement and a thick slab at the inside for a low resistance.
Outer Low Doping	The pn junction is built by low doped regions on the ring. This is the doping for the outside, e.g. low p-doping.
Inner Low Doping	The pn junction is built by low doped regions on the ring. This is the doping for the inside, e.g. low n-doping.
Outer Hi Doping	Further away from the ring, the doping concentration is increased to lower the contact resistance. This is the doping for the outside, e.g. high p-doping.
Inner Hi Doping	Further away from the ring, the doping concentration is increased to lower the contact resistance. This is the doping for the inside, e.g. high n-doping.
Outer Contact Doping	To form ohmic contacts the doping concentration is further increased. This is the contact doping for the outside contacts, e.g. p++.
Inner Contact Doping	To form ohmic contacts the doping concentration is further increased. This is the contact doping for the outside contacts, e.g. n++.
Metal 1	Definition of the lower metal layer, next to the silicon structures.
Metal 1 Perforation Blocking	Some processes do automatic perforation of wide metal structures. This layer marks the areas for the lower metal layer, where this should be prevented to improve RF performance.
Metal 2	Definition of the upper metal layer, next to the top surface.

Metal 2 Perforation Blocking	Some processes do automatic perforation of wide metal structures. This layer marks the areas for the upper metal layer, where this should be prevented to improve RF performance.
Heater	This layer defines the heater structure. The heater is often made from a different metal than metal 1 with a higher resistance.
Heater Contact	This layer marks the overlap between the heater layer and metal 1 as contact between both.
Vias Metal 1 – Metal 2	This layer defines the vias between the lower and the upper metal layer. The shape of the vias and the via pattern is defined in the Metallization tab.
Silicon Contact Plugs	This layer defines the contact plugs between the silicon and the lower metal layer. The shape of the contact plugs and their pattern is defined in the Doping tab.

Drawing tab:

In this tab you can choose, what should be drawn in the graph window. Having all on results in a rather unclear and confusing graph, but it's all shown. Switching any switch results in a recalculation of the geometry (sorry), but it shouldn't take too long to see the new graph.

Cheers,

Marc Schneider

marc.schneider@kit.edu